

Week 1: EEG Basics and Batch Processing

Adaptive EEG Quality Control and Feature Database

Supervisor: Prof. Dr. Mohammed Aquil Mirza, COMP
Student Mentor: Mr. Shihao WANG

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Learning outcomes

- Explain EEG as multi-channel voltage time series, not “brain reading”.
- Describe eyes-open / eyes-closed conditions and the alpha-band example.
- Trace the Task 2 pipeline from raw files to 2-second windows.
- Design a safe batch loop with logging and per-subject error handling.

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Background

Why this course?

- Task 2 scales a small EEG analysis into a dataset-level feature platform.
- The target is not clinical diagnosis; it is reliable data engineering for EEG features.
- It turns EEG analysis into reliable, repeatable data engineering.

EEG in one picture

- EEG records tiny voltage changes from sensors placed on the scalp.
- Think of EEG as many synchronized time series: channel $\times$ time.
- Artifacts such as movement, eye blinks, muscle tension, or bad electrodes can dominate the signal.



From Task 1 to Task 2

The Task 1 starting point

- Process one LEMON subject.
- Separate Eyes Open and Eyes Closed segments.
- Compute power spectral density (PSD).
- Show alpha-blockade: alpha power often changes when eyes close/open.

The Task 2 target

- Scan the full dataset folder.
- Split data into 2-second windows.
- Reject bad windows before feature extraction.
- Store features and quality decisions in PostgreSQL.

Why 2-second windows?

- A long recording is easier to audit when divided into short windows.
- Window-level QC lets us reject only contaminated segments instead of discarding an entire subject.
- Each feature row can link back to a specific window.



Batch Processing

Batch processing requirements

- Never crash the whole run because one subject is corrupted.
- Log success, failure reason, runtime, and input path for every subject/session.
- Use stable IDs: `subject_id`, `session_id`, `window_id`, `run_id`.

Minimal batch-loop pattern

- `discover_files(raw_root)`
- for each subject/session: `try process_one()`
- except: write `processing_errors` and continue
- at the end: write `processing_summary.csv`

Common failure cases

- Missing or unreadable EEG file.
- Unexpected channel names or sampling rate.
- No valid eyes-open/eyes-closed annotation.
- Memory or timeout issues for large recordings.



Deliverable

Week 1 deliverable

- Reusable functions from the Task 1 script.
- Folder scanner and batch runner.
- Processing summary with successful and failed subjects.
- Short README explaining how to run the batch pipeline.